Complete Time Series Analysis & Forecasting Master Notes

Integrated from Lecture 5 (ARIMA/SARIMA) and Lecture 6 (Forecasting Models)

SECTION 1: FULL INTEGRATED NOTES

1. Time Series Components: Trend, Seasonality, Cyclical, Random. Additive Model: Y_t = Trend + Seasonal + Random. Multiplicative Model: Y_t = Trend × Seasonal × Random. 2. Forecasting Types: Ex Ante – true future forecasts. Ex Post – in-sample validation forecasts. 3. Forecasting Models: Naïve: y■(T+h) = yT. Regression: Linear and Nonlinear models. EWMA: $EMA_t = \alpha x_t + (1-\alpha)EMA_{t-1}$. AR(p), MA(q), ARMA(p,q), ARIMA(p,d,q). SARIMA: $(p,d,q)(P,D,Q)_m$. 4. ARIMA Identification: AR(p): PACF cuts off, ACF tails. MA(q): ACF cuts off, PACF tails. ARMA: both tail. White Noise: no spikes. 5. Differencing: Regular differencing removes trend. Seasonal differencing: $x_t - x_{t-S}$. 6. Model Evaluation: AIC / BIC – lower is better. Ljung-Box test: want $p > 0.05$. Residuals should resemble white noise. 7. Bias-Variance Tradeoff: Underfitting = High Bias. Overfitting = High Variance. 8. Workflow: Plot → Transform (log if needed) → Difference → Check ACF/PACF → Fit → Check residuals → Compare AIC → Forecast.

SECTION 2: CONDENSED EXAM CHEAT SHEET

AR(p): PACF cuts off after p. MA(q): ACF cuts off after q. ARMA: both ACF and PACF tail off. ARIMA(p,d,q): p = AR terms d = differencing q = MA terms SARIMA(p,d,q)(P,D,Q)m: m = seasonal period (12 monthly, 4 quarterly) Seasonal MA: ACF spikes at m, 2m. Seasonal AR: PACF spikes at m, 2m. Ljung-Box: H0: No autocorrelation. Good model → $p > 0.05$. AIC/BIC: Lower is better, penalises complexity.

SECTION 3: EXAM-STYLE PRACTICE QUESTIONS WITH SOLUTIONS

Q1: ACF cuts off after lag 2, PACF tails. What model? Solution: MA(2). Q2: PACF cuts off after lag 3, ACF tails. Solution: AR(3). Q3: Significant spikes at lags 12 and 24 in PACF. Solution: Seasonal AR(2) with m=12. Q4: Ljung-Box $p = 0.01$. Interpretation? Solution: Residual autocorrelation remains. Model inadequate. Q5: Why seasonal differencing? Solution: Removes repeating seasonal mean shifts.

SECTION 4: ACF/PACF PATTERN RECOGNITION GUIDE

AR Model: ACF → gradual decay PACF → sharp cutoff MA Model: ACF → sharp cutoff PACF → gradual decay ARMA: Both decay gradually. White Noise: No significant spikes. Seasonal AR: PACF spikes at seasonal lags. Seasonal MA: ACF spikes at seasonal lags.

SECTION 5: ARIMA/SARIMA DECISION FLOWCHART

Step 1: Plot data. Step 2: Trend present? → Difference (d=1). Step 3: Seasonality present? → Seasonal difference (D=1). Step 4: Examine ACF/PACF. Step 5: Identify AR or MA patterns. Step 6: Fit candidate models. Step 7: Compare AIC/BIC. Step 8: Perform Ljung-Box test. Step 9: If $p > 0.05$ → Accept. Step 10: Produce forecasts with prediction intervals.